

Green Coffee Quality Standards & Roasting QA Manual

Meridian Roasters Co. Document reference: MR-QA-STD-002 Version: 2.3 Effective: 1 September 2025 Owner: Head of Quality Applies to: green coffee intake, lot release, blend composition, roast production, finished-goods release Review cycle: annual

1. Purpose

This manual defines the quality standards that green coffee must meet to be accepted, the specifications each product must be built to, the conditions under which a component may be substituted, and the records that must exist before finished goods are released for sale.

It is written for the quality team, the roasting team and production planning. Where it conflicts with a production schedule, this manual takes precedence and the schedule moves.

2. Cupping protocol

2.1 Method

All green coffee is evaluated on arrival using the SCA cupping protocol, ten cups per lot, three assessors, scored independently before discussion. Scores are recorded to one decimal place against the lot record.

Where assessors disagree by more than two points, a fourth assessor is brought in and the lot is rescored. The original scores are retained; they are not overwritten.

2.2 Score bands

Band	Score	Disposition
Exceptional	88.0 and above	Single-origin release; micro-lot programme eligible
Speciality	84.0 to 87.9	Single-origin release; blend base eligible
Premium	82.0 to 83.9	Blend component only; not released as single origin
Commercial	80.0 to 81.9	Blend filler only, maximum thirty per cent of a blend by weight
Reject	Below 80.0	Not released. Quarantine and supplier claim under MR-PROC-HB-003

§7

2.3 Score is a property of the lot, not the supplier

A supplier who consistently delivers at 86 has not thereby earned an 86 on an unscored lot. Every lot is scored on arrival. No lot enters production on the basis of the supplier's history.

3. Product specifications

Each product has a written specification. Production against a specification that has not been met is a deviation and is handled under §5.

3.1 Cascadia Blend (SKU MR-BLD-001)

The company's highest-volume product and the base of most subscription plans.

Attribute	Specification
Base component	Colombian washed, 60% by weight
Secondary component	Guatemalan washed, 25% by weight
Balance	Brazilian natural, 15% by weight
Minimum cupping score, base component	84.0
Minimum cupping score, secondary	83.0
Minimum cupping score, balance	80.0
Roast profile	Medium, first crack plus 1:45 to 2:10
Target finished cup	Balanced, sweet, cocoa and stone fruit, medium acidity

The Colombian base is what customers recognise as Cascadia. It carries the sweetness and the acidity structure. Substituting the base changes the character of the cup more than any other single change available to us, and subscription customers who have drunk it for months will notice within one bag.

3.2 Other products

SKU	Product	Base origin	Min base score
MR-BLD-002	Nightwatch Blend	Sumatra	82.0
MR-BLD-003	Harbour Espresso	Brazilian natural	82.0
MR-SO-011	Yirgacheffe Kochere	Ethiopian washed	86.0
MR-SO-012	Huila Reserva	Colombian washed	86.0
MR-SO-013	Sumatra Lintong	Sumatra wet-hulled	84.0
MR-SO-014	Cerrado Amarelo	Brazilian natural	84.0
MR-DEC-001	Quietwater Decaf	Colombian, EA process	83.0

Single-origin products may not be substituted at all. If the named origin is unavailable, the product goes out of stock. This is not negotiable and has been tested twice.

4. Lot release

A green lot may enter production only when all of the following exist on the lot record:

- a cupping score from three assessors
- a moisture reading between 10.0 and 12.0 per cent
- a defect count within grade
- an intake date and a linked purchase order
- no open supplier claim against the lot

A lot that has been in the silo more than nine months from intake must be re-cupped before use. Green coffee ages. A lot that scored 84 on arrival may not score 84 a year later, and an aged lot used at its original score is a specification failure hiding behind a stale record.

5. Substitution policy

This is the most consequential section in this manual and the one most often applied under time pressure.

5.1 When substitution may be considered

Substitution of a blend component may be considered only when:

- the specified component is genuinely unavailable, not merely inconvenient to retrieve, and
- the alternative component meets the minimum score for the position it is filling, and
- the substitution is recorded on the batch record before the roast, not after

Production pressure is not a justification. Running out of a product is an acceptable outcome and is always preferable to shipping something the customer did not order under a name they recognise.

5.2 Authority

Substitution	Authority
Within the same origin, score delta 1.0 or less	Roast lead
Cross-origin, score delta 2.0 or less	QA Lead
Cross-origin, score delta above 2.0	QA Lead and Head of Quality, jointly
Any substitution affecting a blend's named base component	Head of Quality, in writing
Any substitution on a single-origin product	Not permitted under any authority

5.3 Score delta

Score delta is the difference between the cupping score of the specified component and the cupping score of the substitute. It is calculated against the **specification minimum**, not against the score of the lot that happened to be in use last week.

Example. Cascadia Blend base carries a specification minimum of 84.0. A substitute lot scoring 79.5 gives a delta of 4.5, which is above the 2.0 threshold, requires joint QA Lead and Head of Quality authority under §5.2, and triggers the notification obligation in §5.4.

A substitute scoring below 80.0 is in the Reject band under §2.2 and may not be used in any blend regardless of authority. A substitute between 80.0 and 81.9 is Commercial band and may not exceed thirty per cent of the blend by weight, which disqualifies it from any base position.

5.4 Customer notification obligation

Where a substitution produces a score delta greater than 2.0 points against specification, affected customers must be notified before or with despatch.

The notification must:

- state that the product differs from its usual composition
- state the nature of the difference in plain terms
- offer the customer the option to decline the shipment or to substitute a different product at no charge
- be issued to every customer whose order contains the affected batch, including wholesale accounts

The obligation sits with the Head of Quality to raise and with Customer Experience to execute. It is not discretionary and it is not waived by a judgement that the cup is "probably fine". A customer who discovers the substitution by tasting it has been denied a choice they were entitled to make.

Failure to notify where notification was required is a quality incident in its own right, independent of whether any customer complains, and must be raised as such.

5.5 Batch record requirements

Every substitution must be recorded on the batch record with:

- the specified component and the substitute used
- both cupping scores and the calculated delta
- the authorising individual and the authority level applied
- whether notification was required under §5.4, and if so, the date it issued
- the reason the specified component was unavailable

A QA note on the batch record that describes a substitution but does not record the authority, the delta and the notification status is incomplete and is a finding in its own right at audit.

6. QA note taxonomy

Batch records carry free-text QA notes. Notes are classified as follows and should be written so the class is unambiguous.

Class	Meaning	Action
Observation	A characteristic worth recording, no deviation	None
Deviation	Production departed from specification	Root cause required within five working days
Concession	A deviation accepted under named authority	Authority must be recorded
Hold	Batch not released pending investigation	Blocks despatch

A note describing a substitution and a flatter cup profile is a **deviation**, and if the batch was released it is a **concession**, which means an authority must be recorded against it. A batch released with a deviation note and no recorded authority is a control failure whether or not the coffee was acceptable.

7. Finished-goods release

Finished goods are released when:

- the batch record is complete, including any substitution record under §5.5
- a post-roast cup has been taken and scored against the product's target profile
- any notification obligation under §5.4 has been discharged, or explicitly recorded as not applicable with a reason

Release is a positive act. A batch that nobody blocked has not been released; it has been overlooked.

8. Roast profile control

Each product has a target profile expressed as time from first crack. Profiles are held in the roasting system and are not adjusted to compensate for green quality. Roasting a flatter lot darker to mask the difference is expressly prohibited: it changes the product the customer receives while concealing the reason, and it defeats the notification obligation in §5.4 by removing the evidence a cupper would use.

Where a substitute component genuinely requires a different approach, that is a new product specification and goes through §3, not through an undocumented profile adjustment on the day.

9. Traceability

Every finished bag must be traceable to its roast batch, every roast batch to its green lot, every green lot to its purchase order, and every purchase order to its supplier. The chain is:

```
order line -> roast batch -> green lot -> purchase order -> supplier
```

This chain exists so that a quality complaint can be resolved against evidence rather than against opinion. Where the chain is broken at any hop, the complaint cannot be attributed, and under MR-CC-POL-004 §7.1 an unattributable quality complaint is treated as company-attributable.

Traceability records are retained for three years.

10. Known risks

Recorded here because they have materialised before.

Single-origin dependency on the Colombian base. Cascadia Blend's base component has one qualified supplier. Any interruption to that supply forces a choice between a substitution above the §5.3 threshold and a stock-out. There is no third option, and pretending otherwise under pressure is how §5.4 gets missed. Qualification of a second Colombian source has been on the procurement plan since 2024.

Aged balance-component lots. Brazilian natural lots held for the balance position tend to sit longest in the silo and are the most likely to be reached for in a shortage. They are also the most likely to have aged past their recorded score. See §4 on re-cupping at nine months.

Substitution under despatch pressure. Substitution decisions are made on production mornings when a roast is scheduled and the silo is short. That is precisely when the authority requirements in §5.2 and the notification obligation in §5.4 are most likely to be skipped, because both require someone outside the roastery to be available.

11. Revision history

Version	Date	Change
1.0	Mar 2021	Initial issue
2.0	Feb 2023	Score bands aligned to SCA; single-origin substitution prohibited
2.1	Nov 2023	Nine-month re-cupping requirement added
2.2	Apr 2024	QA note taxonomy introduced
2.3	Sep 2025	§5.4 customer notification obligation added following INC-2024-11; §5.5 batch record requirements expanded